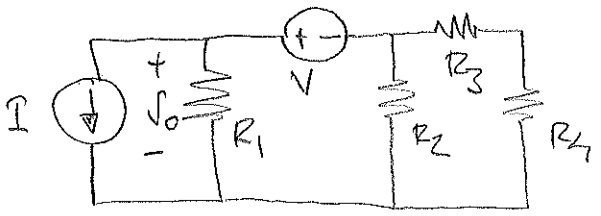
Esobe

Lösningsförslag

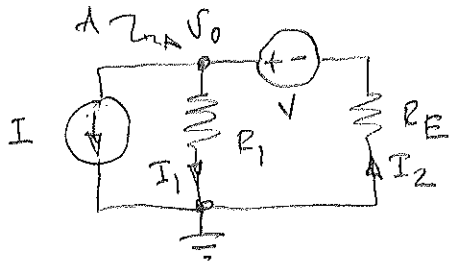
2021-01-14

1.



$V = 25\,V$
$R_1 = 50\,\Omega$
$R_2 = 150\,\Omega$
$R_3 = 20\,\Omega$
$R_4 = 55\,\Omega$

NODSPÄNNINGSANALYS FÖR ATT BERÄKNA SPÄNNINGEN V_0 SAMT ELEKTRISK EFFEKT SOM STRÖMKÄLLAN LEVERERAR.



ANGÄTT NODSP. V_0 OCH JORD.

$$R_E = \left[\frac{1}{150} + \frac{1}{75}\right]^{-1} = 50\,\Omega \qquad (1)$$

FÖR NOD A GÄLLER KSTL

$$-I + I_2 - I_1 = 0 \qquad (2)$$

DÄR $I_1 = \dfrac{V_0}{R_1}$ (3)

KSPL FÖR HÖGRA DELEN GER

$$-R_E I_2 + V - V_0 = 0$$

$$I_2 = \frac{V - V_0}{R_E} \qquad (4)$$

(4) & (3) I (2) GER

$$-I + \frac{V - V_0}{R_E} - \frac{V_0}{R_1} = 0$$

$$\frac{V}{R_E} - I = V_0\left(\frac{1}{R_E} + \frac{1}{R_1}\right)$$

$$V_0 = \frac{\frac{V}{R_E} - I}{\frac{1}{R_E} + \frac{1}{R_1}}$$

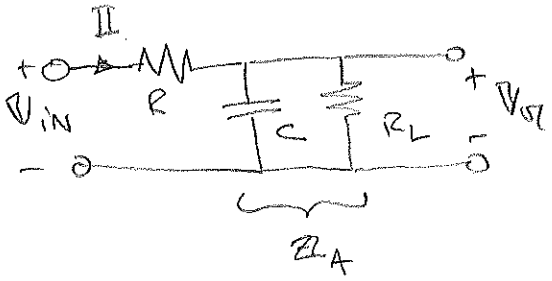
NUMERISKT:

$$V_0 = \frac{\frac{25}{50} - 2}{\frac{1}{50} + \frac{1}{50}} = -37.5 \text{ VOLT}$$

OCH EFFEKTEN BLIR $P = -V_0 I = 37.5 \cdot 2 = 75$ WATT

SVAR: $V_0 = -37.5\,V$
$P = 75\,W$

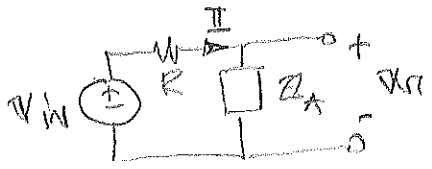
2.



$R = 160\,\Omega$

$C = 5\,nF$

ΚΥΚΛΩΜΑΤΙΚΑ



ΚΥΚΛΟΦΟΡΙΑΣ

$$\begin{cases} -V_{IN} + (R + Z_A)\,I = 0 & (1) \\ -V_{OUT} + Z_A I = 0 & (2) \end{cases}$$

(2) ΚΑΙ (1) ΔΙΝΕΙ

$$-V_{IN} + (R + Z_A)\frac{V_{OUT}}{Z_A} = 0$$

$$\frac{V_{OUT}}{V_{IN}} = \frac{Z_A}{R + Z_A} \qquad (3)$$

ΟΠΟΥ $Z_A = \left[\frac{1}{R_L} + j\omega C\right]^{-1} = \frac{R_L}{1 + j\omega C R_L}$ (4)

(4) Χ (3) ΔΙΝΕΙ

$$H(\omega) = \frac{V_{OUT}}{V_{IN}} = \frac{\dfrac{R_L}{1 + j\omega C R_L}}{\left(R + \dfrac{R_L}{1 + j\omega C R_L}\right)(1 + j\omega C R_L)} =$$

$$= \frac{R_L}{R(1 + j\omega C R_L) + R_L}$$

ΤΟ ΠΛΑΤΟΣ ΤΗΣ $H(\omega)$

$$|H(\omega)| = \frac{R_L}{|R + R_L + j\omega C R_L R|} =$$

$$= \frac{R_L}{\sqrt{(R + R_L)^2 + (\omega C R_L R)^2}}$$

FÖR SMÅ ω, DVS $\omega \to 0$ FÅS MED DIAGRAMMET

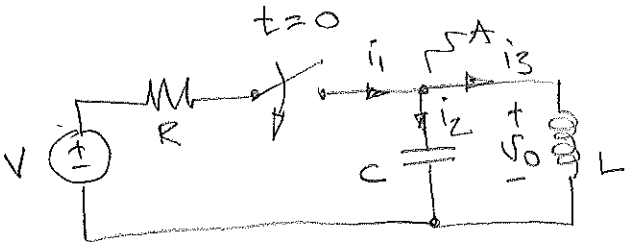
$$|H(\omega)| \lim_{\omega \to 0} = \frac{R_L}{R+R_L} = \frac{1}{2} \Rightarrow R_L = R = 160\,\Omega$$

SVAR: a)

$$H(\omega) = \frac{R_L}{R_L + R + j\omega C R_L R}$$

b) $R_L = 160\,\Omega$

3)



$V = 75\,V$
$C = 6.25\,nF$
$L = 250\,mH$
$R = 125\,\Omega$

KSTL för NOD A ger

$$i_1 - i_2 - i_3 = 0 \qquad (1)$$

DÄR

$$\begin{cases} i_1 = \dfrac{V - v_0}{R} \\ i_2 = C\dfrac{dv_0}{dt} \\ i_3 = i_3(0) + \dfrac{1}{L}\int v_0\,dt \end{cases} \qquad (2)$$

($i_3(0) = 0$)

(2) i (1) ger

$$\frac{V - v_0}{R} - C\frac{dv_0}{dt} - \frac{1}{L}\int v_0\,dt = 0 \qquad (2.5)$$

DERIVERA MAP t

$$-\frac{1}{R}\frac{dv_0}{dt} - C\frac{d^2v_0}{dt^2} - \frac{1}{L}v_0 = 0$$

ELLER SOM

$$\frac{d^2v_0}{dt^2} + \frac{1}{RC}\frac{dv_0}{dt} + \frac{1}{LC}v_0 = 0 \qquad (3)$$

($\frac{1}{RC} = 2\alpha$, $\frac{1}{LC} = \omega_0^2$)

VI IDENTIFIERAR

$$\begin{cases} 2\alpha = \dfrac{1}{RC} \Rightarrow \alpha = \dfrac{1}{2RC} \\ \omega_0^2 = \dfrac{1}{LC} \Rightarrow \omega_0 = \dfrac{1}{\sqrt{LC}} \end{cases}$$

NUMERISKT

$$\begin{cases} \alpha = \dfrac{1}{2 \cdot 125 \cdot 6.25 \cdot 10^{-6}} = 640 \\ \omega_0 = \dfrac{1}{\sqrt{250 \cdot 10^{-3} \cdot 6.25 \cdot 10^{-6}}} = 800 \end{cases} \qquad (4)$$

Se ▽

$\omega_0 > \alpha$, ALTSÅ ÄR LÖSNINGARNA UNDERDÄMPADE

$$\begin{cases} v_0(t) = e^{-\alpha t}(A\cos(\beta t) + B\sin(\beta t)) \\ \beta = \sqrt{\omega_0^2 - \alpha^2} \end{cases} \qquad (5)$$

MEN $v_0(0) = 0$ (KOND. ÄR EN SPÄNNINGSTRÖG) SÅ (5) BLIR

$$v_0(0) = A \Rightarrow A = 0$$

B DÅ? ANVÄND (2.5) VID $t = 0$

(2.5) VID $t = 0$: $\frac{V}{R} - \underbrace{\frac{v_0}{R}}_{=0} - C\frac{dv_0}{dt}\Big|_{t=0} - \frac{1}{L}\underbrace{\int_0^0 v_0\,dt}_{=0} = 0$

$$\Rightarrow \frac{V}{R} - C\frac{dv_0}{dt}\Big|_{t=0} = 0 \qquad (6)$$

(5) I (6) GER

$$\frac{V}{R} - C\frac{d}{dt}\left(Be^{-\alpha t}\sin(\beta t)\right) = 0$$

$$\frac{V}{R} - CB\left(-\alpha e^{-\alpha t}\sin(\beta t) + e^{-\alpha t}\beta(-\cos(\beta t))\right)\Big|_{t=0} = 0$$

$$\frac{V}{R} + CB\beta = 0$$

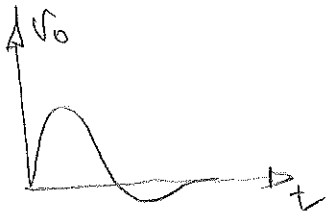
$$B = \frac{V}{RC\beta}$$

NUMERISKT

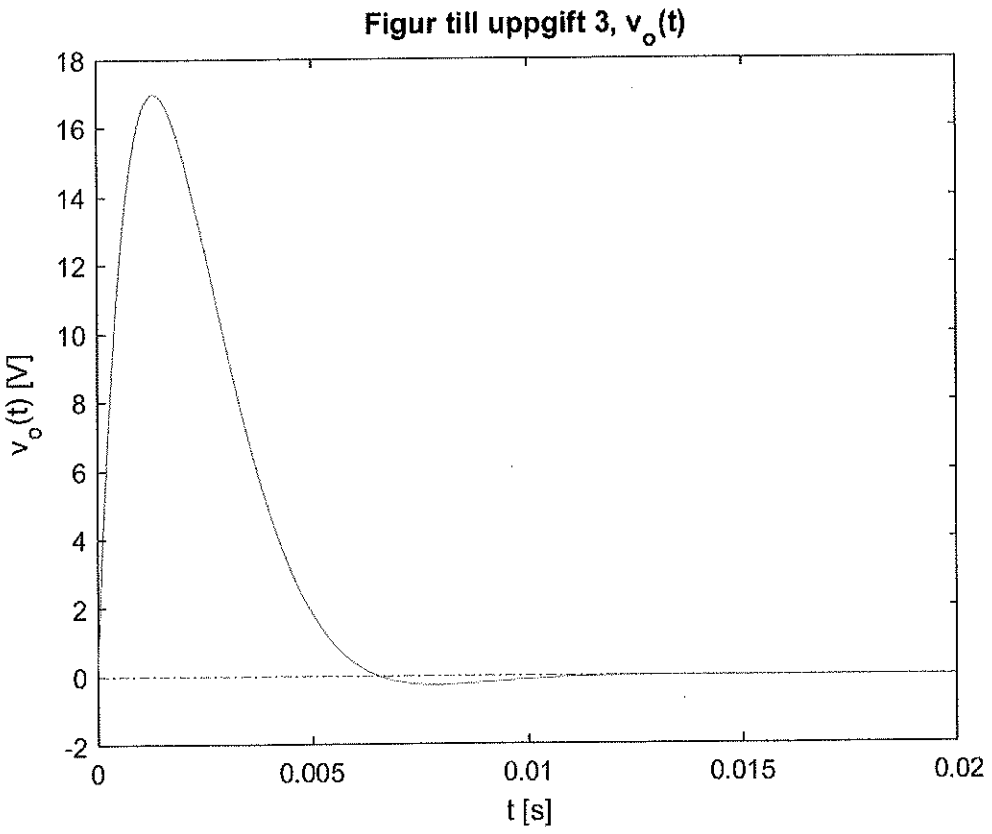
$$B = \frac{75}{125 \cdot 6.25 \cdot 10^{-6} \cdot 480} = \frac{200}{3}$$

SVAR: SPÄNNINGEN ÖVER SPOLEN BLIR

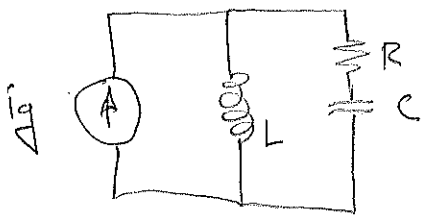
$$v_0(t) = \frac{200}{3}e^{-640t}\sin(480t)$$



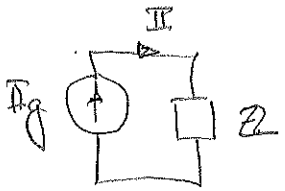
SE FIGUR →


Figur till uppgift 3, $v_o(t)$
18
16
14
12
10
8
6
4
2
0
-2
$v_o(t)$ [V]
0
0.005
0.01
0.015
0.02
t [s]

4)



HOPPATILL KOMPLEXA TALPLANET



VI VET ATT $i_g(t) = 80\cos(1250t)$ mA

KOMPLEX EFFEKT $ KAN TA SKRIVAS

$$\$ = \frac{1}{2}VI^* = \frac{1}{2}Z\,II^* = \frac{1}{2}Z|I|^2 \qquad (1)$$

IMPEDANSEN FÖR KRETSEN

$$Z^{-1} = \frac{1}{j\omega L} + \frac{1}{R+\frac{1}{j\omega C}} =$$

$$= \frac{1}{j\omega L} + \frac{j\omega C}{1+j\omega CR} = \frac{1+j\omega CR - \omega^2 LC}{j\omega L(1+j\omega CR)}$$

BLIR SOM

$$Z = \frac{j\omega L - \omega^2 LCR}{1-\omega^2 Lc + j\omega CR} \qquad (2)$$

VI FÅR NU EFFEKTEN

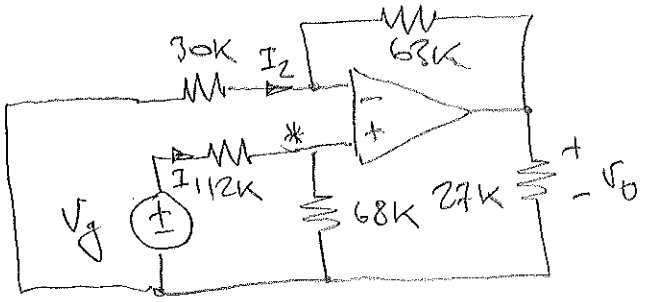
$$\$ = \frac{1}{2}\,\frac{(j1250\cdot 16 - 1250^2\cdot 16\cdot 40\cdot 10^{-9}\cdot 10\cdot 10^3)(80\cdot 10^{-3})^2}{1-1250^2\cdot 16\cdot 40\cdot 10^{-9} + j1250\cdot 40\cdot 10^{-9}\cdot 10\cdot 10^3} =$$

$$= \frac{1}{2}\,\frac{j20000 - 10000}{1-1+j\frac{1}{2}}(80\cdot 10^{-3})^2 =$$

$$= \frac{11180.3\,\angle 116.56^\circ}{\frac{1}{2}\angle 90^\circ}(80\cdot 10^{-3})^2 \simeq 143.1\,\angle 26.56^\circ \text{ VA}$$

SVAR: $\$ \simeq 143.1\,\angle 26.56^\circ$ VA

5)



BESTÄMMER SAMBANDEN MELLAN V_g OCH V_0

KSPL GER $\quad -V_g + (12K + 68K) I_1 = 0$

$$I_1 = \frac{V_g}{80K} \qquad (A)$$

SPÄNNINGEN VID * BLIR

$$V_* = 68K \cdot I_1 = \frac{68}{80} V_g \qquad (B)$$

KSPL I DEN YTTRE SLINGAN GER

$$-V_0 - 63K \cdot I_2 + V_* = 0 \qquad (C)$$

OCH

$$30K \cdot I_2 + V_* = 0 \Rightarrow I_2 = -\frac{V_*}{30K} \qquad (D)$$

(D) I (C)

$$-V_0 + 63K \frac{V_*}{30K} + V_* = 0$$

$$V_0 = 3.1 V_* \overset{(B)}{=} 3.1 \frac{68}{80} V_g \simeq 2.635 V_g$$

DVS

$$V_0 \simeq 2.635 V_g$$

$$V_g \simeq 0.3797 V_0$$

OM $-12 \leq V_0 \leq 12 \Rightarrow -4.55 \leq V_g \leq 4.55$ V

SVAR: $-4.55 \leq V_g \leq 4.55$ V

OM $-12 \leq V_0 \leq 12$ VOLT